

Requirements Table (PES1UG24CS411)

PROBLEM STATEMENT #42 | Developer Tools & IT Operations

Internal Microservice Catalog & Health Portal

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ping each registered microservice's health endpoint every 30 seconds and compute a rolling 24-hour availability percentage.	High	Pass: 3 consecutive failed probes trigger an alert. Fail: unreachable service still shown as healthy.	Core monitoring capability the portal is built around.
FR-002	Functional	The system shall allow a DevOps Engineer to register a new microservice with its endpoint URL, owning team, and dependency list.	High	Pass: service appears in catalog immediately with correct metadata. Fail: duplicate endpoint accepted silently.	Catalog can't function without a registration path.
FR-003	Functional	The system shall let a System Architect view an interactive dependency graph of service-to-service call relationships.	High	Pass: clicking a node highlights its direct upstream/downstream dependencies. Fail: graph fails to load or mislabels an edge.	Needed to reason about blast radius during incidents.
FR-004	Functional	The system shall aggregate and display each service's	Medium	Pass: docs refresh within 5 minutes of a spec change. Fail:	Cuts context-switching for

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		OpenAPI/Swagger documentation in a unified viewer.		stale docs served past that window.	engineers integrating with a service.
FR-005	Functional	The system shall allow a DevOps Engineer to configure per-service alert rules (channel, threshold, quiet hours).	High	Pass: alert delivered to the configured channel within 60s of a threshold breach. Fail: alert sent to a default channel regardless of config.	Without this, FR-001's alerting is one-size-fits-all and gets muted.
NFR-001	Nonfunctional	The system shall render the dependency graph for architectures with up to 200 services without noticeable UI lag.	High	Pass: node click/hover response stays under 200ms with 200 services loaded. Fail: UI visibly stutters or freezes.	Given constraint — graph is the centerpiece of FR-003.
NFR-002	Nonfunctional	The health-check subsystem itself shall maintain 99.9% uptime and recover missed probes after a restart.	High	Pass: scheduler backfills missed checks within 2 minutes of recovery. Fail: silent gap in the 24-hour availability window.	A monitor that goes down during an incident defeats its purpose.

UML DIAGRAM

